

RETAIL BANKING TRANSACTION ANALYSIS

Python Data Analysis + Power BI Business Intelligence

Industry-level project documentation based on the supplied Python notebooks, cleaned datasets and Power BI report.

Data scope: 2,377,169 transaction records × 8 columns

Power BI report pages: Customer, Account, Transaction and Final

1. Executive Summary

This project is an end-to-end retail banking analytics solution that combines Python for data preparation and exploratory analysis with Microsoft Power BI for semantic modelling, KPI reporting and interactive business analysis.

The documentation has been aligned directly with the supplied Power BI report. The report contains four analytical pages — Customer, Account, Transaction and Final — and uses customer, account, transaction-code and date dimensions around the high-volume account-transactions fact table.

The transaction fact table has a confirmed shape of **2,377,169 rows × 8 columns**. Its schema includes transaction_id, account_id, linked_transaction_id, created_time, transaction_type, transaction_code, amount and channel/year reporting fields as shown in the supplied data screenshot.

What this project demonstrates

- Data understanding and cleaning with Python.
- Business-rule based validation instead of blind outlier deletion.
- Customer and account segmentation.
- Power BI semantic modelling and reusable KPI measures.
- Executive dashboard design with slicers, KPI cards and drill-down analytical pages.
- Translation of operational banking data into business questions and recommendations.

2. Data Landscape

Dataset inventory

Table	Role	Shape / scope
account_transactions_cleaned	Transaction fact	2,377,169 × 8
bank_accounts_cleaned	Account portfolio	3,553 × 9
customer_profiles_cleaned	Customer dimension	1,000 × 7
customer_names_mapping	Customer-name mapping	1,000 × 2
transaction_codes_cleaned	Transaction reference	15 × 5
DateTable	Power BI date dimension	Used in report for fiscal and calendar analysis

Transaction fact-table schema

Column	Role in analysis
transaction_id	Transaction identifier; used by Power BI for transaction count
account_id	Account associated with each transaction
linked_transaction_id	Optional linkage between related transactions
created_time	Transaction timestamp
transaction_type	Credit / Debit classification
transaction_code	Transaction-code reference
amount	Transaction monetary value
channel	Transaction channel; used by Power BI channel analysis

3. Python Data Preparation & Analysis

The supplied notebooks provide the technical preparation layer before the data reaches Power BI. The cleaning workflow covers data understanding, missing-value inspection, duplicate checks, date handling, validity checks, outlier investigation and feature engineering.

Feature engineering visible in the supplied analytical workflow

- Customer age and age_group are created for customer segmentation.
- Customer-level balance is available for value-oriented analysis.
- Transaction amount distribution is investigated using skewness and percentile-based analysis.
- The analysis avoids automatically deleting legitimate high-value financial transactions simply because they are statistical extremes.

The Python layer should be treated as the data-quality and analytical foundation; Power BI then converts those prepared fields into a reusable semantic model and management dashboard.

4. Power BI Dashboard Architecture

The supplied PBIX was inspected at the report-layout level. It contains four report pages and a dedicated executive page. The documentation below describes the actual visual structure found in the report rather than replacing it with a generic dashboard description.

Page	Purpose	Dashboard elements identified
Customer	Customer profile and value analysis	Customer count card; age-group chart; Top 10 States; Top 10 cities; Top 10 Customers; net balance
Account	Account and credit portfolio analysis	Account-type mix; account-status mix; Savings/CASA balance; FD/TD balance; outstanding loans; lo
Transaction	Transaction behaviour analysis	Transaction count; transaction value by Credit/Debit; transaction value by channel; transaction-count
Final	Executive management dashboard	8 KPI cards; channel/account/customer/location/age/date/transaction-type slicers; Top 10 Customers

5. Power BI — Customer Page

The Customer page is designed around customer scale, segmentation and relationship value. The page does not simply show a customer count; it combines demographic distribution, geography and net-balance concentration.

Visuals identified in the PBIX

- Count of Customers — card visual based on customer_id.
- Age-group distribution — column chart using age_group and customer count.
- Top 10 States — clustered bar chart using state and customer count.
- Top 10 Cities — clustered bar chart using city and customer count.
- Top 10 Customers — customer_name against net_balance.
- Net Balance by Age Group — pie chart.
- Net Balance by State — clustered bar chart.
- Total Net Balance and Average Net Balance — card visuals.

Business interpretation

The Customer page is primarily a **relationship-value and segmentation view**. It allows an analyst to move from customer volume to where customers are concentrated and then to where net balance is concentrated. This is the correct place to discuss customer value, geographic concentration and age-based portfolio composition.

6. Power BI — Account Page

The Account page shifts from customer relationships to product and credit portfolio analysis. Its visuals use `bank_accounts_cleaned` and `customer_profiles_cleaned` fields and are therefore directly tied to the account layer of the model.

KPI cards identified

- Savings Balance / CASA Balance.
- FD Balance / TD Balance.
- Outstanding Loan Amount.
- Loan Amount / Loan Disbursements.
- Net Balance measure.
- Transaction-linked account count.

Portfolio visuals

- Account type distribution.
- Account status distribution.
- Loan amount by customer age group.
- Balance by account type.
- Loan amount versus interest rate.
- Account count across interest-rate bands.
- Account count across loan term months.

This page is the project's main **portfolio and credit analytics layer**. It supports funding-side analysis through Savings and FD balances and credit-side analysis through loan amount, outstanding loans, interest rates and term structure.

7. Power BI — Transaction Page

The Transaction page is the strongest direct use of the 2,377,169-row transaction fact table. The page uses `transaction_id`, `amount`, `transaction_type`, `channel`, `transaction_code`, `DateTable` fields and `Transaction_Amt_Band`.

Transaction visuals identified

- Transaction Count — card using `transaction_id`.
- Transaction Value by Credit/Debit — donut chart using `transaction_type` and sum of amount.
- Transaction Value by Channel — clustered column chart using `channel` and sum of amount.
- Transaction Count over Time — line chart using `Date` and transaction count.
- Transaction Volume by Transaction Code — clustered bar chart.
- Quarter-Wise Trend of Transaction Amount — column chart using `Fiscal Year` and `Fiscal Quarter`.
- Transaction Amount Distribution — clustered bar chart using `Transaction_Amt_Band` and transaction count.
- Transaction Volume by Weekday — clustered bar chart using `weekday` and transaction count.

Transaction data scale

2,377,169 rows × 8 columns are present in the transaction table. This makes it the high-volume fact layer of the project and justifies the use of aggregation, date dimensions and amount-band segmentation in Power BI.

The documentation deliberately does not invent channel totals, quarterly amounts or transaction-code rankings that were not extracted as numeric result values from the PBIX. The dashboard design itself is documented exactly so that the published repository remains auditable.

8. Power BI — Final Executive Dashboard

The Final page is the executive layer of the project. It consolidates the most important measures and allows users to filter the dashboard without leaving the management view.

Eight KPI cards identified in the Final page

KPI label	Source / field used by the PBIX
CUSTOMER COUNT	customer_profiles_cleaned → Count of customer_id
NO OF ACCOUNTS	account_transactions_cleaned → Count of account_id
NET BALANCE	bank_accounts_cleaned → net_balance_measure
TRANSACTION COUNT	account_transactions_cleaned → Count of transaction_id
CASA BALANCE	bank_accounts_cleaned → savings_balance
TD BALANCE	bank_accounts_cleaned → fd_balance
LOAN DISBURSEMENTS	bank_accounts_cleaned → Sum of loan_amount
OUTSTANDING LOANS	bank_accounts_cleaned → outstanding_loan_amt

Interactive controls and visuals

- Slicers: channel, account type, customer name, state/city, age group, date and transaction type.
- Top 10 Customers by net balance.
- Net Balance by age group.
- Count of Accounts by account status.
- Quarterly Transaction Value Trend.
- Transaction Volume by amount band.
- State/city/customer summary table with customer count and net-balance contribution.

Dashboard quality note: the Final page's 'NO OF ACCOUNTS' card is currently configured in the PBIX as Count of account_id from the transaction fact table. That is not the same as a distinct account count. For an industry-standard KPI, consider changing it to `DISTINCTCOUNT(bank_accounts_cleaned[account_id])` and keeping the current transaction-level count under a clearly named transaction-linked account metric.

9. Dashboard-Led Business Insights

The following findings are intentionally written from the structure and measures of the Power BI dashboard rather than from unsupported external calculations.

- Customer value is a core analytical theme: the dashboard ranks the Top 10 Customers by net balance and also distributes net balance by age group and state.
- The account dashboard connects product mix with financial value through Savings, FD, Loan and Net Balance measures.
- The transaction dashboard is designed to compare value and volume separately, which is important because high transaction counts do not necessarily imply high transaction value.
- Transaction behaviour can be analysed simultaneously by channel, transaction type, transaction code, amount band, weekday and fiscal quarter.
- The Final dashboard brings customer, account and transaction measures into a single executive view and provides slicers for rapid segmentation.
- The presence of a DateTable enables consistent fiscal-year and fiscal-quarter analysis instead of relying only on raw transaction timestamps.

10. KPI Governance & Modelling Observations

Recommended naming / modelling improvements before public portfolio publication

- Rename the Final-page 'NO OF ACCOUNTS' card or change its aggregation to DISTINCTCOUNT of bank account IDs.
- Use explicit measure names such as [Customer Count], [Transaction Count], [Net Balance], [CASA Balance], [TD Balance], [Loan Disbursements] and [Outstanding Loans].
- Keep transaction-level measures sourced from account_transactions_cleaned and portfolio measures sourced from bank_accounts_cleaned.
- Use the DateTable consistently for all transaction time-series visuals.
- Document the definition of net balance so users understand the business meaning behind the measure.

These recommendations do not change the supplied dashboard; they are portfolio-quality governance suggestions based on the actual Power BI configuration inspected for this project.

11. Recommended GitHub Repository Structure

Retail-Banking-Python-PowerBI/

■■■ README.md

■■■ dataset/

■ ■■■ account_transactions_cleaned.csv # 2,377,169 × 8

■ ■■■ bank_accounts_cleaned.csv

■ ■■■ customer_profiles_cleaned.csv

■ ■■■ customer_names_mapping.csv

■ ■■■ transaction_codes_cleaned.csv

■■■ python/

■ ■■■ 01_Cleaning.ipynb

■ ■■■ 02_Analysis.ipynb

■■■ powerbi/

■ ■■■ Retail_Banking_Analysis.pbix

■■■ documentation/

■ ■■■ Project_Documentation.pdf

■■■ presentation/

■ ■■■ Project_Presentation.pptx

■■■ screenshots/

■■■ dashboard.png

Large-file recommendation

Because the transaction table contains 2.37 million rows, avoid treating the CSV as an ordinary small repository file. Use Git LFS, a documented external storage location, or a reproducible sample plus a clear note explaining how the full dataset is obtained.

12. Conclusion

This project demonstrates an end-to-end banking analytics workflow in which Python provides the data-quality and analytical foundation and Power BI provides the business-facing reporting layer. The most important improvement in this revised documentation is that the narrative is now explicitly anchored to the supplied Power BI report: each page, KPI, slicer and major visual is documented according to the actual PBIX configuration.

With 2,377,169 transaction records, the project has sufficient scale to demonstrate practical BI skills beyond a small classroom dataset. The Customer page addresses relationship value, the Account page addresses portfolio and credit structure, the Transaction page addresses high-volume behavioural analysis, and the Final page brings these domains together for executive decision support.

Before publishing the repository as a polished portfolio project, the recommended KPI naming correction and documentation of DAX measure definitions should be completed. This will make the solution more transparent and closer to an industry reporting standard.

Technology Stack

Python: Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Business Intelligence: Microsoft Power BI

Data: CSV / relational analytical tables

Version control: Git / GitHub

Core methods: Data cleaning, validation, feature engineering, EDA, semantic modelling, KPI design and dashboarding